FREIRE DE ALBUQUERQUE João Pedro

Dual-degree engineering student – UFRN (Brazil) / Toulouse INP

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EDUCATION

TOULOUSE INP | *Dual-degree program* 2026 – 2028
- Electronic Engineering – Department 3EA (Electronics, Electrical Energy and Automation)

FEDERAL UNIVERSITY OF RIO GRANDE DO NORTE (UFRN) | *Brazil* 2024 – 2028
- Bachelor's in Electrical Engineering

EXPERIENCE

IEEE INDUSTRIAL ELECTRONICS SOCIETY UFRN, Brazil
President of the IEEE IES Student Chapter at UFRN 2026
- Coordinating national and international events
- Leading the automation and AI teams and organizing short courses
- Grew the chapter by 95% (from 19 to 37 members)

ATLAS EXPERIMENT (CERN) – TILECAL CALORIMETER UFRN, Brazil
Research Intern Sep 2024 – Aug 2025
- Project: implementation of synthesized logic on FPGA for convolutional neural networks (CNNs) in the TileCal of the ATLAS experiment
- Developed a CNN in Python, synthesized on FPGA with hls4ml, for low-latency classification of hadron collision products in the ATLAS experiment (CERN), in order to reduce the effect of *pile-up*

SKILLS

Technical OrCAD, Keysight ADS, Ansys HFSS, Python, C++

Languages Portuguese: native • English: fluent • French: intermediate

Interests Microstrip antennas, signal processing, artificial intelligence, applied electromagnetics, defense

PUBLICATIONS

NoticIEEEro IEEE Region 9 International Magazine

AWARDS

POTIGUAR INNOVATION CHALLENGE IN AUTOMATION AND ELECTRONICS Quatro Ponto Zero, private company
- Designed a smart scale integrated with a web interface for a local food business